Part II. Nonlinear Control Systems Design. pg. 191

- Part I : analyze behavior of given nonlinear system.
In Part II, we design the nonlinear control system.
"construct a feedback control law to make the system display the desired behavior".

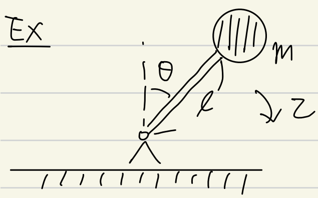
nonlinear regulation, nonlinear tracking, methods for designing...

① Stabilization problem

Given a nonlinear dynamic system $\dot{x} = f(x, u, t)$,
find a control law u s.t. for $\forall x_0 \in \Omega$, $x \xrightarrow{t \to \infty} 0$.

* If we want $x \to x_0$, then consider $x - x_0$ as the state.

Ex

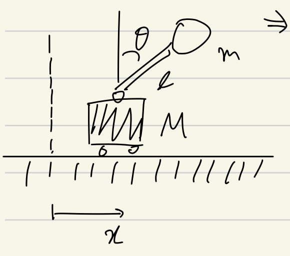


$$J\ddot{\theta} - mgl\sin\theta = \tau.$$

$$\tau = -k_d\dot{\theta} - k_p\theta - mgl\sin\theta$$

$$\Rightarrow J\ddot{\theta} + k_d\dot{\theta} + k_p\theta = 0.$$

$$\Rightarrow \theta \to 0 \quad \text{for suitable } k_d, k_p > 0.$$



$$(M+m)\ddot{x} + ml\cos\theta\,\ddot{\theta} - ml\sin\theta\,\dot{\theta}^2 = u$$

$$m\ddot{x}\cos\theta + ml\ddot{\theta} - mg\sin\theta = 0.$$

: Very difficult.

② Asymptotic Tracking Problem.

$$\begin{cases} \dot{x} = f(x, u, t) \\ y = h(x) \end{cases}$$

Desired output : y_d. ($y_d^{(k)}$: bdd)

Find u s.t. $\forall x_0 \in \Omega$, $\begin{cases} |y(t) - y_d(t)| \xrightarrow{t \to \infty} 0 \\ |x| : \text{bdd}. \end{cases}$

$y(t) \equiv y_d(t)$ $\forall t \geq 0$ is possible : perfect tracking

Normally, tracking problems are more difficult than stabilization problems.

$y(t) \to y_d(t) \iff e(t) := y(t) - y_d(t)$

$\ddot{y} + f(\dot{y}, y, u) = 0 \Rightarrow \ddot{e} + f(\dot{e}, e, u, y_d, \dot{y}_d, \ddot{y}_d) = 0$

tracking related stabilizer

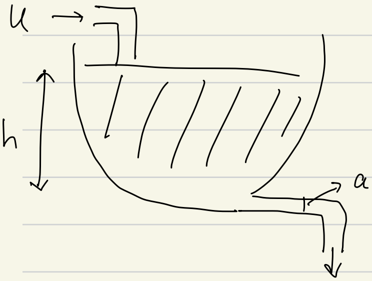
✓ desired behavior
- stability
- accuracy (tracking)
- robustness (withstand disturbances)
- cost (actuators, sensors, computers ...)

Methods of Nonlinear Control Design.
- Trial and error
- Feedback linearization — Ch 6
- Robust control — Ch 7
- Adaptive control — Ch 8
- Gain-scheduling

Ch 6. Feedback Linearization

§6.1 Intuitive concepts

6.1.1 Feedback Linearization and the canonical form



$A(h)$: cross section of the tank.

$$\frac{d}{dt}\left(\int_0^h A(s)\,ds\right) = u(t) - a\sqrt{2gh}$$

$$A(h) \cdot \dot{h} = u - a\sqrt{2gh}$$

$$u(t) = a\sqrt{2gh} + A(h)\,v$$

$$\Rightarrow \dot{h} = v$$

$\tilde{h} = h(t) - h_d$

$v = -\alpha \tilde{h} \Rightarrow \dot{h} + \alpha\tilde{h} = \dot{\tilde{h}} + \alpha\tilde{h} = 0 \Rightarrow \tilde{h} \to 0$ as $t \to \infty$

If $h_d = h_d(t)$, let $v = \dot{h}_d(t) - \alpha\tilde{h}$, $\alpha > 0$: constant

$\Rightarrow \dot{\tilde{h}} + \dot{h}_d = \dot{h}_d - \alpha\tilde{h} \Rightarrow \dot{\tilde{h}} + \alpha\tilde{h} = 0$, $\tilde{h} \to 0$ as $t \to \infty$.

idea: cancel nonlinearities

$x^{(n)} = f(x) + b(x)u$ $\quad u = \frac{1}{b}(v - f) \Rightarrow x^{(n)} = v$.

$$\dot{X} = \begin{pmatrix} x_1 \\ \vdots \\ x_{n-1} \\ x_n \end{pmatrix}' = \begin{pmatrix} x_2 \\ \vdots \\ x_n \\ f(x) + b(x)u \end{pmatrix}$$

$v = -k_0 x - k_1 \dot{x} - \cdots - k_{n-1} x^{(n-1)}$,

give suitable k_i so that

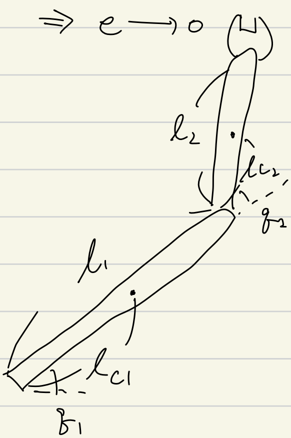
$p^n + k_{n-1} p^{n-1} + \cdots + k_1 p_1 + k_0 = 0$

has all roots are strictly in the left-half complex plane.

$\Rightarrow x^{(n)} + k_{n-1} x^{(n-1)} + \cdots + k_0 x = 0$, $x \xrightarrow[t\to\infty]{} 0$

$v = x_d^{(n)} - k_0 e - k_1 \dot{e} - \cdots - k_{n-1} e^{(n-1)}$, $e = x(t) - x_d(t)$

$\Rightarrow e \longrightarrow 0$



$$\begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix} \begin{pmatrix} \ddot{q}_1 \\ \ddot{q}_2 \end{pmatrix} + \begin{pmatrix} -h\dot{q}_2 & -h\dot{q}_1 - h\dot{q}_2 \\ h\dot{q}_1 & 0 \end{pmatrix} \begin{pmatrix} \dot{q}_1 \\ \dot{q}_2 \end{pmatrix} + \begin{pmatrix} g_1 \\ g_2 \end{pmatrix} = \begin{pmatrix} \tau_1 \\ \tau_2 \end{pmatrix}$$

$H_{11} = m_1 l_{c1}^2 + m_2 (l_1^2 + l_{c2}^2 + 2 l_1 l_{c2} \cos q_2) + I_2$

H_{22}, $H_{12} = H_{21}$, h, g_1, g_2

$$H(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = \tau.$$

H is invertible.

$$\tau = H(q)\,v + C(q,\dot q)\,\dot q + g(q),$$

$$v = \ddot q_d - 2\lambda \dot{\tilde q} - \lambda^2 \tilde q, \quad \tilde q = q - q_d, \quad \lambda > 0$$

$$\Rightarrow \ddot{\tilde q} + 2\lambda \dot{\tilde q} + \lambda^2 \tilde q = 0, \quad \tilde q \xrightarrow{t\to\infty} 0$$

6.1.2 Input-State Linearization

$\dot x = f(x,u)$ ① $z = z(x)$, $u = u(x,v)$ $\Rightarrow$ $\dot z = Az + bv$ ② Design v.

state transition (for $z = z(x)$); input transition (for $u = u(x,v)$)

<u>Ex</u>

$$\dot x_1 = -2x_1 + a x_2 + \sin x_1$$
$$\dot x_2 = -x_2 \cos x_1 + \underline{u} \cos(2x_1)$$

$$\left\{\begin{aligned} z_1 &= x_1, \\ z_2 &= a x_2 + \sin x_1. \end{aligned}\right\} \text{ state trans}$$

$$\Rightarrow \dot z_1 = -2z_1 + z_2$$

$$\dot z_2 = a(-x_2 \cos x_1 + u\cos(2x_1)) + \cos x_1 \cdot (-2x_1 + a x_2 + \sin x_1)$$
$$= -2z_1 \cos z_1 + \cos z_1 \sin z_1 + a u \cos(2z_1)$$

input trans

$$u = \frac{1}{a\cos(2z_1)}\left(v - \cos z_1 \sin z_1 + 2z_1 \cos z_1\right) \Rightarrow \left\{\begin{aligned} \dot z_1 &= -2z_1 + z_2 \\ \dot z_2 &= v \end{aligned}\right.$$

$$v = -2z_2 \Rightarrow \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}' = \begin{pmatrix} -2 & 1 \\ 0 & -2 \end{pmatrix}\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} : \; z_1, z_2 \xrightarrow{t\to\infty} 0$$

$$\therefore \; u = \frac{1}{a\cos(2x_1)}\left(-2a x_2 - 2\sin x_1 - \cos x_1 \sin x_1 + 2x_1 \cos x_1\right),$$

then $x_1 = z_1 \to 0$, $x_2 = (z_2 - \sin z_1)/a \to 0$ as $t \to \infty$.

<u>Rmk</u> Not global (not well defined when $x_1 = \frac{\pi}{4} \pm \frac{k\pi}{2}$, $k \in \mathbb{Z}$)

6.1.3 Input - Output Linearization

$$\begin{cases} \dot{x} = f(x,u) \\ y = h(x) \end{cases} \qquad y(t) \to y_d(t)$$

$$\begin{cases} \dot{x}_1 = \sin x_2 + (x_2+1)\, x_3 \\ \dot{x}_2 = x_1^5 + x_3 \\ \dot{x}_3 = x_1^2 + u \\ y = x_1 \end{cases}$$

$$\ddot{y} = (x_2+1)u + \underbrace{(x_1^5+x_3)(x_3+\cos x_2) + (x_2+1)\, x_1^2}_{= f_1(x)}$$

$$u = \frac{1}{x_2+1}(v - f_1) \Rightarrow \ddot{y} = v.$$

$$v = \ddot{y}_d - k_1 e - k_2 \dot{e} \Rightarrow \ddot{e} + k_2 \dot{e} + k_1 e = 0 \quad \cdots (*)$$

$(k_1, k_2 > 0)$ $\qquad$ $e \to 0$

Rmk Not defined at $x_2 = -1$

Are we done? System is 3rd order, but $(*)$ is 2nd order.

$$\dot{x}_3 = x_1^2 + \frac{1}{x_2+1}(\ddot{y}_d - k_1 e - k_2 \dot{e} + f_1)$$

x_3 should be stable

: This part of the system became "unobservable" in the input-output system.

$\rightsquigarrow$ "Internal Dynamics"

Ex
$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}' = \begin{pmatrix} x_2^3 + u \\ u \end{pmatrix} \qquad y = x_1$$

$$\dot{y} = x_2^3 + u$$

$$u = -x_2^3 - e(t) + \dot{y}_d(t)$$

$$\Rightarrow \dot{e} + e = 0.$$

$$\Rightarrow \dot{x}_2 + x_2^3 = \dot{y}_d - e, \qquad |\dot{y}_d - e| \le D.$$

$$x_2 > D^{1/3} \Rightarrow \dot{x}_2 < 0$$

$$x_2 < -D^{1/3} \Rightarrow \dot{x}_2 > 0$$

$\Rightarrow$ stable.

Usually, it is very difficult to determine the stability of internal dynamics.

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}' = \begin{pmatrix} x_2 + u \\ u \end{pmatrix} \quad y = x_1. \quad y \to y_d$$

$\dot{y} = x_2 + u \qquad \underline{u = -x_2 + \dot{y}_d - e} \quad \Rightarrow \quad \dot{e} + e = 0.$

internal dynamics : $\dot{x}_2 + x_2 = \dot{y}_d - e \quad \Rightarrow x_2$: bdd

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}' = \begin{pmatrix} x_2 + u \\ -u \end{pmatrix} \quad y = x_1 \qquad \dot{y} = x_2 + u \quad \underline{u = -x_2 + \dot{y}_d - e}$$

$\Rightarrow \dot{e} + e = 0.$

internal dynamics : $\dot{x}_2 - x_2 = -\dot{y}_d + e$ ∴ x_2 can go to ∞.

∴ not a suitable tracking controller.

* Transfer functions (pg 130)

$$\begin{aligned} \dot{x} &= Ax + bu \\ y &= c^T x \end{aligned} \quad \Rightarrow \quad h(p) = c^T (pI - A)^{-1} b$$

$$\dot{x} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} u, \quad y = \begin{pmatrix} 1 \\ 0 \end{pmatrix}^T x$$

$$\Rightarrow h_1(p) = (1\ 0) \begin{pmatrix} p & -1 \\ 0 & p \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = (1\ 0)\ \frac{1}{p^2} \begin{pmatrix} p & 1 \\ 0 & p \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$h_2(p) = \frac{p-1}{p^2} \qquad = \frac{1}{p^2} (p\ 1) \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{p+1}{p^2}$$

For linear systems,
✓ internal dynamics is stable if zeros of transfer function are in the left half plane.

pf : 222 ~ 223

6.2 Mathematical Tools

$f : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ vector field.

The Jacobian of f is denoted by ∇f, $(\nabla f)_{ij} = \frac{\nabla f_i}{\partial x_j}$

Def 6.1 Suppose $h : \mathbb{R}^n \longrightarrow \mathbb{R}$, $f : \mathbb{R}^n \longrightarrow \mathbb{R}^n$ smooth.

Lie derivative of h with respect to f = $L_f h = \nabla h\, f$

= directional derivative of h along the direction of vector f.

$$L_f^i h = L_f^i (L_f^{i-1} h)$$

ex $\dot{x} = f(x)$ single output system

$y = h(x)$. $\quad \dot{y} = \frac{\partial h}{\partial x} \dot{x} = L_f h \qquad \ddot{y} = \frac{\partial (L_f h)}{\partial x} \dot{x} = L_f^2 h$.

Def 6.2 $f.g$ = vector fields. Lie bracket of $f.g$ = $[f.g]$ ← vector field.

$= \nabla g f - \nabla f g$

$$f(x) = \begin{pmatrix} -2x_1 + ax_2 + \sin x_1 \\ -x_2 \cos x_1 \end{pmatrix} \qquad g(x) = \begin{pmatrix} 0 \\ \cos 2x_1 \end{pmatrix}$$

$$\Rightarrow [f.g] = \begin{pmatrix} 0 & 0 \\ -2\sin(2x_1) & 0 \end{pmatrix} \begin{pmatrix} -2x_1 + ax_2 + \sin x_1 \\ -x_2 \cos x_1 \end{pmatrix}$$

$$- \begin{pmatrix} -2 + \cos x_1 & a \\ x_2 \sin x_1 & -\cos x_1 \end{pmatrix} \begin{pmatrix} 0 \\ \cos 2x_1 \end{pmatrix}$$

$$= \begin{pmatrix} a \cos 2x_1 \\ \cos x_1 \cos 2x_1 - 2\sin(2x_1)(-2x_1 + ax_2 + \sin x_1) \end{pmatrix}$$

Lemma 6.1 (i) $[\alpha_1 f_1 + \alpha_2 f_2 . g] = \alpha_1 [f_1 . g] + \alpha_2 [f_2, g]$

$[f, \alpha_1 g_1 + \alpha_2 g_2] = \alpha_1 [f . g_1] + \alpha_2 [f . g_2]$

(ii) $[g.f] = -[f.g]$

(iii) h: smooth ftn $L_{[f,g]}h = L_f L_g h - L_g L_f h$.

$$L_{[f,g]}h = \nabla h\,[f,g] = \frac{\partial h}{\partial x}\left(\frac{\partial g}{\partial x} f - \frac{\partial f}{\partial x} g\right)$$

$$L_f L_g h - L_g L_f h = \nabla(L_g h) f - \nabla(L_f h) g = \nabla\left(\frac{\partial h}{\partial x} g\right) f - \nabla\left(\frac{\partial h}{\partial x} f\right) g$$

$$= \left(\frac{\partial h}{\partial x}\frac{\partial g}{\partial x} + g^T \frac{\partial^2 h}{\partial^2 x}\right) f - \left(\frac{\partial h}{\partial x}\frac{\partial f}{\partial x} + f^T \frac{\partial^2 h}{\partial x^2}\right) g$$

$$= \frac{\partial h}{\partial x}\left(\frac{\partial g}{\partial x} f - \frac{\partial f}{\partial x} g\right)$$

Def 6.3 $\phi : \Omega \to \mathbb{R}^n$ is a diffeomorphism if ϕ: smooth. $\exists \phi^{-1}$, ϕ^{-1}: smooth.

Lem 6.2 $\phi : \Omega \to \mathbb{R}^n$ smooth. $\nabla\phi$: non-singular at $x = x_0 \in \Omega$.
$\Rightarrow \phi$ defines a diffeomorphism near x_0.

Ex state transformation $\begin{pmatrix} z_1 \\ z_2 \end{pmatrix} = \phi(x) = \begin{pmatrix} 2x_1 + 5x_1 x_2^2 \\ 3\sin x_2 \end{pmatrix}$

$$\frac{\partial \phi}{\partial x} = \begin{pmatrix} 2 + 5x_2^2 & 10x_1 x_2 \\ 0 & 3\cos x_2 \end{pmatrix}$$

$\exists$ smooth inverse in.

$\Omega = \{(x_1, x_2) : |x_2| < \frac{\pi}{2}\}$

Frobenius Theorem

$h(x_1, x_2, x_3)$: unknown

$$\begin{cases} \frac{\partial h}{\partial x_1} f_1 + \frac{\partial h}{\partial x_2} f_2 + \frac{\partial h}{\partial x_3} f_3 = 0 \\ \frac{\partial h}{\partial x_1} g_1 + \frac{\partial h}{\partial x_2} g_2 + \frac{\partial h}{\partial x_3} g_3 = 0 \end{cases}$$

: $\exists$ solution h?

For what kind of f & g?

Def 6.4 A linearly independent set of vector fields $\{f_1, f_2, \cdots, f_m\}$ on $\mathbb{R}^n$ is completely integrable if $\exists\, n-m$ scalar functions $h_1, \cdots, h_{n-m}$ satisfying $\nabla h_i f_j = 0$, $1 \le i \le n-m$, $1 \le j \le m$.

$m(n-m)$ PDEs.

Def 6.5 A linearly independent set of vector fields $\{f_1, f_2, \cdots, f_m\}$ is <u>involutive</u> if $\exists \alpha_{ijk} : \mathbb{R}^n \to \mathbb{R}$ s.t.

$$[f_i, f_j](x) = \sum_{k=1}^{m} \alpha_{ijk}(x) f_k(x) \qquad \forall i, j.$$

Thm 6.1 (Frobenius)

$\{f_1, f_2, \cdots f_m\}$: linearly independent vector fields.

The set is completely integrable $\iff$ involutive.

Ex

$$(*)\begin{cases} 4x_3 \dfrac{\partial h}{\partial x_1} - \dfrac{\partial h}{\partial x_2} = 0 \\ -x_1 \dfrac{\partial h}{\partial x_1} + (x_3^2 - 3x_2)\dfrac{\partial h}{\partial x_2} + 2x_3 \dfrac{\partial h}{\partial x_3} = 0. \end{cases}$$

$$f_1 = \begin{pmatrix} 4x_3 \\ -1 \\ 0 \end{pmatrix} \qquad f_2 = \begin{pmatrix} -x_1 \\ x_3^2 - 3x_2 \\ 2x_3 \end{pmatrix}$$

$$[f_1, f_2] = \nabla f_2 f_1 - \nabla f_1 f_2 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -3 & 2x_3 \\ 0 & 0 & 2 \end{pmatrix}\begin{pmatrix} 4x_3 \\ -1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} -x_1 \\ x_3^2 - 3x_2 \\ 2x_3 \end{pmatrix}$$

$$= \begin{pmatrix} -4x_3 \\ 3 \\ 0 \end{pmatrix} - \begin{pmatrix} 8x_3 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -12x_3 \\ 3 \\ 0 \end{pmatrix} = -3f_1 + 0 f_2$$

$\therefore \exists$ solution h of $(*)$.